

# Harsh Agarwal

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## Profile

Third-year CSE student at JIET with proven track record in full-stack development, IoT automation, and AI integration. Hackathon winner with expertise in C, Python, Java, and modern web technologies. Experienced in API integration, Database Management, and college event anchoring. Combines strong technical skills with effective communication to deliver innovative solutions.

## Projects

### LitWise

- An AI-powered book recommendation platform leveraging machine learning algorithms.
- Next.js, FastAPI, and Goodreads dataset. K-Means clustering, TF-IDF cosine similarity.

### BlogWave

- A full stack blogging platform enabling user to create, share, explore diverse content and community engagement.
- Create and managing blog posts, while ensuring robust user authentication, data security
- Node.js, Express.js, MySQL, Multer, etc.

## Experience

### Literate

- An AI-powered, full-stack reading application designed for enhanced comprehension and focus.
- Enables users to upload books and interact with a smart assistant that answers context-aware questions.
- Retrieval-Augmented Generation (RAG) pipeline, Next.js, TypeScript, Google Genkit, and Firestore.

### Youtube SideView

- A Chrome extension that improves YouTube user experience by implementing a dual-pane interface with resizable side panels for video descriptions and comments.
- Content switching using JavaScript, DOM manipulation and Chrome Extension APIs.

### Unified AI Tools Hub

- A web application integrating a Photo Editor, Text-to-Speech background music generator and Youtube Video Downloader.
- React, TailwindCSS, Azure Cognitive Services, ImageKit, and MongoDB

### Telegram Based Home Automation

- A home automation system using Telegram as the control interface for remote control of lights, fans, and appliances. Integrated motion-detection for automatic lighting activation, enhancing energy efficiency.
- ESP32 microcontrollers, Relays, Motion Sensor and Telegram Bot.

## Courses

**Data Base Management System**, Indian Institute of Technology Kharagpur

**Introduction to Programming in C**, Indian Institute of Technology Kanpur

**Sustainable Engineering Concepts and Life Cycle Analysis**, Indian Institute of Technology Kharagpur

**Innovation, Business Models and Entrepreneurship**, Indian Institute of Technology Roorkee

## Achievements & Activities

|                                 |                                                                                    |
|---------------------------------|------------------------------------------------------------------------------------|
| <b>Regular Event Anchor</b>     | — College institutional events and programs                                        |
| <b>Innovate 2025</b>            | — 2nd Place — Hackathon organized by MANIT J, Modi University and Thapar Institute |
| <b>Hack-ito 2023</b>            | — 1st Place — College Ideathon for Smart India Hackathon                           |
| <b>Hack-ito 2024</b>            | — 2nd Place — College Ideathon for Smart India Hackathon                           |
| <b>LaunchPad Challenge 2024</b> | — Top 6 — Startup pitching at iStart Jodhpur                                       |

## Skills

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| <ul style="list-style-type: none"><li>• <b>Programming Languages:</b> (C   Python   Java)</li><li>• <b>Data Structures &amp; Algorithms</b></li><li>• <b>Problem Solving</b></li><li>• <b>Machine Learning &amp; Data Science</b></li></ul> | <ul style="list-style-type: none"><li>• <b>Communication Skills:</b></li><li>• <b>IoT</b></li><li>• <b>Web Development</b></li><li>• <b>Problem Solving</b></li></ul> |
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## Education

**B.Tech CSE 3rd Year**, Jodhpur Institute of Engineering and Technology (CGPA: 9.39) **September 2023 – Present** Jodhpur, India  
**Higher Secondary Education**, Adarsh Vidya Mandir Shankar Vidya Peeth(78%) **June 2021 – March 2023** Mount Abu, India